

1. Geometry

Nodes: In the context of geometry, a node is a point defined by coordinates in 3D space. Nodes are fundamental units used to define more complex structures. In the dashboard, sensor data is assigned to each node.

Note: nodes are not visible by default.

Lines: In the geometry, a line is defined by two nodes. The line segment between the nodes can represent the boundary of surfaces. In the dashboard, it can also be used to represent structural components such as beams, trusses, and girders.

Surfaces: A surface is a two-dimensional object in three-dimensional space. In the geometry, surfaces are typically defined by three nodes that form a triangle. These surfaces can be used to visualize components such as floors, ceilings, or bridge decks.

Note: When a component is expected to be deformable, it should be made up of multiple pieces rather than a single rigid shape. Each piece can move relative to the others, allowing the overall shape of the component to change.

1.1. Example: One-Story Building

A one-story building example is presented first. The 3D view of the structure is shown in Figure 1. The floor dimensions are 25 m by 31 m, and the story height is 8 m. The global Cartesian coordinate system, node numbering, and geometry are illustrated in Figure 1. Note that the origin of the coordinate system is located at node #1.

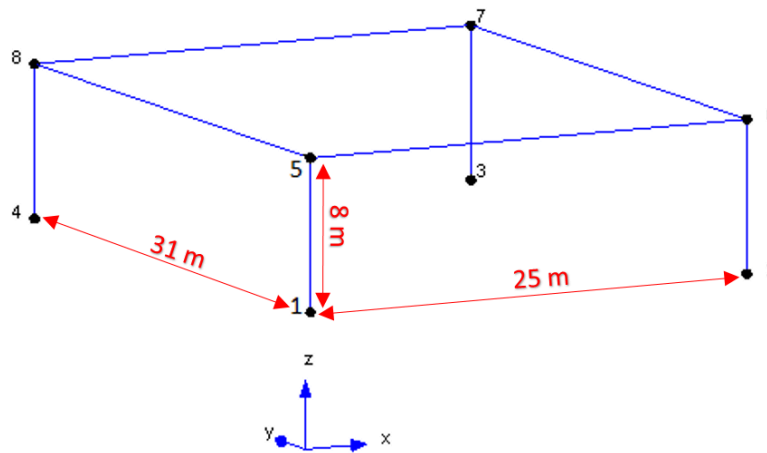


Figure 1: 3D view of the one-story building model

1.1.1. Project Setup

First, create a project by clicking the “+ NEW PROJECT” button in the top right corner of the homepage. Then, select “Larze Ambient Vibration System” from the dropdown menu. This will take users to the project setup page.

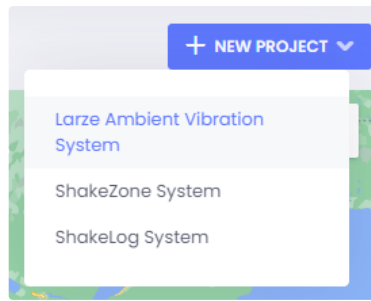


Figure 2: Creating a Larze project

The project setup page includes several sections: **Project Details**, **HAZUS Damage Estimation**, **Location**, **Time Zone**, **Units**, **Geometry**, and **Notes**. To begin, users can name the project in the **Project Details** section, then focus on the **Geometry** section for this example.

In the **Geometry** section, there are multiple steps to define the structure's geometry. Information can be entered manually or uploaded via text files at each step, with data delimited by tabs.

Each node's information should be defined first. The **Nodes** section allows users to specify each node's ID, X, Y, Z coordinates, Sensor ID, and Setup ID. For the one-story building example, the **Nodes** section is shown in Figure 3.

Change Shape ×

ID	X	Y	Z	Sensor ID	Setup ID (0 for Reference Node)	+
1	0	0	0			 
2	25	0	0			 
3	25	31	0			 
4	0	31	0			 
5	0	0	8	5	1	 
6	25	0	8	6	1	 
7	25	31	8	7	1	 
8	0	31	8	8	1	 

Choose File No file chosen

Figure 3: Geometry — Nodes Settings

Next, the lines should be defined. Each line connects two nodes. The line settings for the project are shown in Figure 4.

Change Shape ×

Node_1	Node_2	+	
1	5	 	▲
2	6	 	
3	7	 	
4	8	 	
5	6	 	
6	7	 	
7	8	 	
8	5	 	▼

Choose File

No file chosen

Figure 4: Geometry — Line Settings

Next, the surfaces are defined. The current version of the Dashboard only accepts triangular meshes/surfaces; quadrilateral meshes are not supported at this time. The surface settings for the example are shown in Figure 5.

Change Shape ×

Node_1	Node_2	Node_3	+	
5	6	7	 	▲
5	7	8	 	▼

Choose File

No file chosen

Figure 5: Geometry — Surface Settings

The remaining settings are used for HAZUS estimation and are therefore not covered in this tutorial. Click the “✓” **(Update)** button to save changes. Users can verify the geometry by clicking “**Preview.**” The 3D model of the example is shown in Figure 6. To submit the changes to the servers, users can click the “**Submit**” button at the bottom of the page.

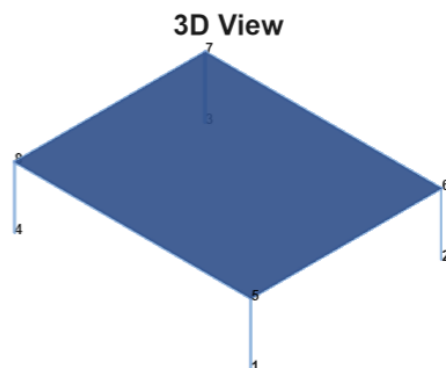


Figure 6: Geometry Preview — 3D Model of the Example

The Geometry section also allows users to configure **“Setup Info.”** Here, **“Setup”** refers to multi-setup tests. This part is skipped in this example since only one setup is involved. It will be discussed further in the 3-story building example.